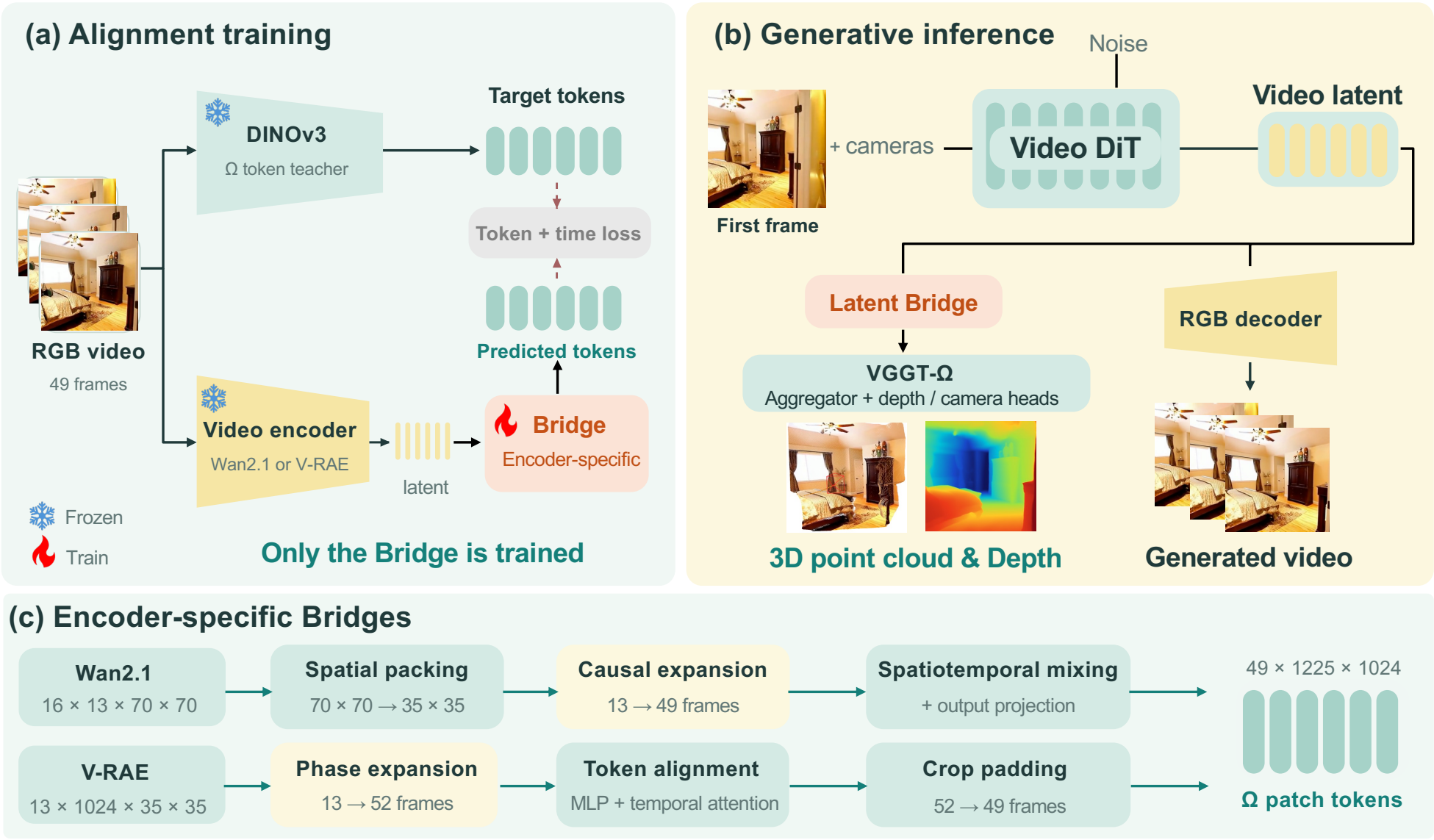

(a) Alignment training
DINOv3
Ω token teacher
Target tokens
Token + time loss
Predicted tokens
RGB video
49 frames
Video encoder
Wan2.1 or V-RAE
latent
Bridge
Encoder-specific
Frozen
Train
Only the Bridge is trained
(b) Generative inference
Noise
First frame
+ cameras
Video DiT
Video latent
Latent Bridge
RGB decoder
VGGT-Ω
Aggregator + depth / camera heads
3D point cloud & Depth
Generated video
(c) Encoder-specific Bridges
Wan2.1
16 × 13 × 70 × 70
Spatial packing
70 × 70 → 35 × 35
Causal expansion
13 → 49 frames
Spatiotemporal mixing
+ output projection
49 × 1225 × 1024
V-RAE
13 × 1024 × 35 × 35
Phase expansion
13 → 52 frames
Token alignment
MLP + temporal attention
Crop padding
52 → 49 frames
Ω patch tokens